Network Traffic Analysis Report (Wireshark)

Overview

This project demonstrates hands-on network traffic analysis using Wireshark. The objective was to identify normal and potentially suspicious activity by analyzing DNS and HTTPS traffic captured from a real environment.

Objectives

• Capture live network traffic • Analyze DNS and HTTPS activity • Identify top communicating IPs • Detect suspicious patterns • Document findings professionally

Tools Used

Wireshark, macOS Terminal, PDF Reporting

Methodology

Traffic was captured on the Wi-Fi interface (en0) while browsing common websites. The capture duration was approximately 3–5 minutes and saved as a .pcap file.

Key Findings

1. DNS Activity: Frequent queries to domains such as pusher.com, reddit.com, and cloudflare.com were observed. This behavior is consistent with normal web application traffic.

2. Network Flow: The local device (192.168.178.21) communicated primarily with the gateway (192.168.178.1), which routed traffic externally. This is expected in NAT-based home networks.

3. HTTPS Traffic: The majority of traffic used port 443 (TLS encryption), indicating secure communication.

4. TCP Reset Packet: A TCP RST packet from 80.77.84.97 was observed, which may indicate a closed connection or interrupted session. This is not inherently malicious but worth noting.

5. Suspicious Activity Assessment: No clear indicators of compromise were detected. Traffic patterns appear consistent with normal browsing behavior.

Analysis Screenshots

network_capture.pcap

tcp.port == 443

No.	Time	Delta	Source	Destination	Protocol	Length	Info
100067	151.025475	0.000040	192.168.178.21	18.64.211.103	TCP	66	62557 → 443 [ACK] Seq=1862 Ack=6160 Win=130816 Len=0 TSval=1912527106 TSecr=662378868
100068	151.057973	0.032498	80.77.84.97	192.168.178.21	TCP	54	443 → 62359 [RST] Seq=4349 Win=0 Len=0
100072	151.083663	0.019860	2606:4700:10::ac42…	2a02:8071:4583:4a6…	TCP	86	443 → 62555 [ACK] Seq=4049 Ack=4563 Win=131072 Len=0 TSval=2920799514 TSecr=214877110
100076	151.092401	0.000427	192.168.178.21	52.215.231.162	TCP	78	62558 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=1129255923 TSecr=0 SACK_PERM
100077	151.130204	0.037803	52.215.231.162	192.168.178.21	TCP	74	443 → 62558 [SYN, ACK] Seq=0 Ack=1 Win=62643 Len=0 MSS=1420 SACK_PERM TSval=1035020405 TSecr=1129255923
100078	151.130317	0.000113	192.168.178.21	52.215.231.162	TCP	66	62558 → 443 [ACK] Seq=1 Ack=1 Win=132352 Len=0 TSval=1129255961 TSecr=1035020405
100079	151.130772	0.000455	192.168.178.21	52.215.231.162	TCP	1474	62558 → 443 [ACK] Seq=1 Ack=1 Win=132352 Len=1408 TSval=1129255961 TSecr=1035020405 [TCP PDU reassembl
100080	151.130786	0.000014	192.168.178.21	52.215.231.162	TLSv1.3	420	Client Hello (SNI=dashboard.stripe.com)
100081	151.166980	0.036194	18.172.112.46	192.168.178.21	TLSv1.2	98	Application Data
100082	151.167121	0.000141	192.168.178.21	18.172.112.46	TCP	66	62010 → 443 [ACK] Seq=541 Ack=1287 Win=2443 Len=0 TSval=3911581177 TSecr=270712628
100083	151.169069	0.001948	52.215.231.162	192.168.178.21	TCP	66	443 → 62558 [ACK] Seq=1 Ack=1763 Win=74752 Len=0 TSval=1035020444 TSecr=1129255961
100084	151.171660	0.002591	52.215.231.162	192.168.178.21	TLSv1.3	1474	Server Hello, Change Cipher Spec, Application Data
100085	151.171663	0.000003	52.215.231.162	192.168.178.21	TLSv1.3	1461	Application Data, Application Data, Application Data
100086	151.171782	0.000119	192.168.178.21	52.215.231.162	TCP	66	62558 → 443 [ACK] Seq=1763 Ack=2804 Win=129536 Len=0 TSval=1129256003 TSecr=1035020445
100087	151.177054	0.005272	192.168.178.21	52.215.231.162	TLSv1.3	146	Change Cipher Spec, Application Data
100088	151.216006	0.038952	52.215.231.162	192.168.178.21	TLSv1.3	145	Application Data
100089	151.216008	0.000002	52.215.231.162	192.168.178.21	TLSv1.3	145	Application Data
100090	151.216009	0.000001	52.215.231.162	192.168.178.21	TLSv1.3	128	Application Data
100091	151.216126	0.000117	192.168.178.21	52.215.231.162	TCP	66	62558 → 443 [ACK] Seq=1843 Ack=3024 Win=130816 Len=0 TSval=1129256047 TSecr=1035020491
100094	151.414351	0.172483	2606:4700:10::ac42…	2a02:8071:4583:4a6…	TCP	1434	443 → 62555 [ACK] Seq=4049 Ack=4563 Win=131072 Len=1348 TSval=2920799844 TSecr=214877110 [TCP PDU reas
100095	151.414354	0.000003	2606:4700:10::ac42…	2a02:8071:4583:4a6…	TLSv1.3	129	Application Data
100096	151.414356	0.000002	2606:4700:10::ac42…	2a02:8071:4583:4a6…	TLSv1.3	1434	Application Data

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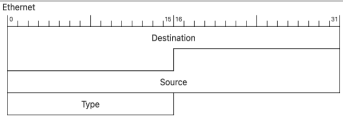
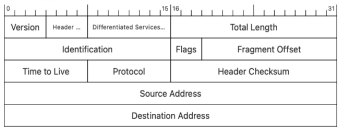
Frame 100076: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits)
Ethernet II, Src: 7e:bc:ad:1a:26:71 (7e:bc:ad:1a:26:71), Dst: AVMAudiovisu_db:3b:c1 (60:b5:8d:db:3b:c1
Internet Protocol Version 4, Src: 192.168.178.21, Dst: 52.215.231.162
Transmission Control Protocol, Src Port: 62558, Dst Port: 443, Seq: 0, Len: 0
    Source Port: 62558
    Destination Port: 443
    [Stream index: 540]
    [Stream Packet Number: 1]
    [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 324237606
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1011 .... = Header Length: 44 bytes (11)
    Flags: 0x002 (SYN)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...0 .... = Acknowledgment: Not set
```

Ethernet



Internet Protocol Version 4



Transmission Control Protocol

This shows the raw value of the sequence number (tcp.seq_raw), 4 bytes

Packets: 111703 · Displayed: 51578 (46.2%)

Profile: Default

network_capture.pcap

dns

	Time	Delta	Source	Destination	Protocol	Length	Info
111241	154.574207	0.000055	192.168.178.21	192.168.178.1	DNS	77	Standard query 0x64ee A ws-mt1.pusher.com
111243	154.577912	0.003636	192.168.178.1	192.168.178.21	DNS	159	Standard query response 0xd48c HTTPS ws-mt1.pusher.com SOA ns-1482.awsdns-57.org
111244	154.579220	0.001308	192.168.178.1	192.168.178.21	DNS	159	Standard query response 0x9a1a AAAA ws-mt1.pusher.com SOA ns-1482.awsdns-57.org
111245	154.580311	0.001091	192.168.178.1	192.168.178.21	DNS	205	Standard query response 0x64ee A ws-mt1.pusher.com A 3.216.92.24 A 52.201.83.248 A 100.52.46.37 A 18.215.1
111503	157.118954	0.135689	192.168.178.21	192.168.178.1	DNS	75	Standard query 0x4bff HTTPS bam.nr-data.net
111504	157.119272	0.000318	192.168.178.21	192.168.178.1	DNS	75	Standard query 0x8b35 AAAA bam.nr-data.net
111505	157.119563	0.000291	192.168.178.21	192.168.178.1	DNS	75	Standard query 0x3afa A bam.nr-data.net
111509	157.126350	0.005622	192.168.178.1	192.168.178.21	DNS	159	Standard query response 0x3afa A bam.nr-data.net CNAME bam.cell.nr-data.net CNAME bam.nr-data.net.cdn.clou
111510	157.126353	0.000003	192.168.178.1	192.168.178.21	DNS	201	Standard query response 0x8b35 AAAA bam.nr-data.net CNAME bam.cell.nr-data.net CNAME bam.nr-data.net.cdn.c
111511	157.133804	0.007451	192.168.178.1	192.168.178.21	DNS	201	Standard query response 0x4bff HTTPS bam.nr-data.net CNAME bam.cell.nr-data.net CNAME bam.nr-data.net.cdn.
111528	158.122528	0.016010	192.168.178.21	192.168.178.1	DNS	80	Standard query 0xf4b5 HTTPS a.nel.cloudflare.com
111529	158.122583	0.000055	192.168.178.21	192.168.178.1	DNS	80	Standard query 0x114d AAAA a.nel.cloudflare.com
111530	158.122622	0.000039	192.168.178.21	192.168.178.1	DNS	80	Standard query 0x692b A a.nel.cloudflare.com
111531	158.122816	0.000194	192.168.178.21	192.168.178.1	DNS	87	Standard query 0x3c97 HTTPS w3-reporting-nel.reddit.com
111532	158.122855	0.000039	192.168.178.21	192.168.178.1	DNS	87	Standard query 0x632b AAAA w3-reporting-nel.reddit.com
111533	158.122886	0.000031	192.168.178.21	192.168.178.1	DNS	87	Standard query 0xa9ff A w3-reporting-nel.reddit.com
111536	158.128354	0.004164	192.168.178.1	192.168.178.21	DNS	108	Standard query response 0x114d AAAA a.nel.cloudflare.com AAAA 2600:1901:0:47fc::
111537	158.128355	0.000001	192.168.178.1	192.168.178.21	DNS	96	Standard query response 0x692b A a.nel.cloudflare.com A 35.190.80.1
111541	158.138914	0.004188	192.168.178.1	192.168.178.21	DNS	131	Standard query response 0xf4b5 HTTPS a.nel.cloudflare.com SOA coleman.ns.cloudflare.com
111543	158.141326	0.002319	192.168.178.1	192.168.178.21	DNS	308	Standard query response 0x3c97 HTTPS w3-reporting-nel.reddit.com CNAME w3-reporting-nel.reddit.com.cdn.clo
111544	158.141845	0.000519	192.168.178.1	192.168.178.21	DNS	203	Standard query response 0x632b AAAA w3-reporting-nel.reddit.com CNAME w3-reporting-nel.reddit.com.cdn.clou
111545	158.142846	0.001001	192.168.178.1	192.168.178.21	DNS	179	Standard query response 0xa9ff A w3-reporting-nel.reddit.com CNAME w3-reporting-nel.reddit.com.cdn.cloudfl

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Frame 111531: Packet, 87 bytes on wire (696 bits), 87 bytes captured (696 bits)
Ethernet II, Src: 7e:bc:ad:1a:26:71 (7e:bc:ad:1a:26:71), Dst: AVMAudiovisu_db:3b:c1 (60:b5:8d:db:3b:c1
Internet Protocol Version 4, Src: 192.168.178.21, Dst: 192.168.178.1
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 73
    Identification: 0x4d70 (19824)
    000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: UDP (17)
    Header Checksum: 0x47cc [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.178.21
    Destination Address: 192.168.178.1
    [Stream index: 25]
User Datagram Protocol, Src Port: 45593, Dst Port: 53
Domain Name System (query)
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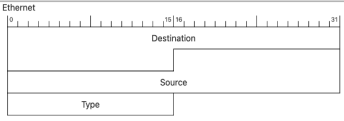
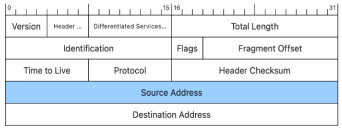
Ethernet



Internet Protocol Version 4

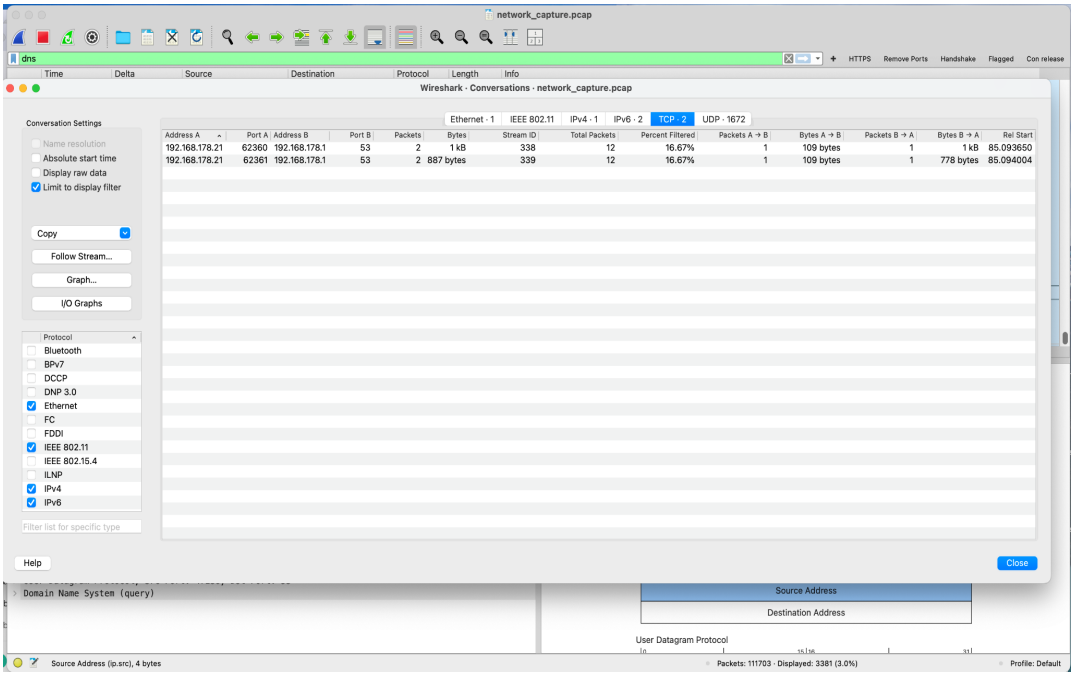


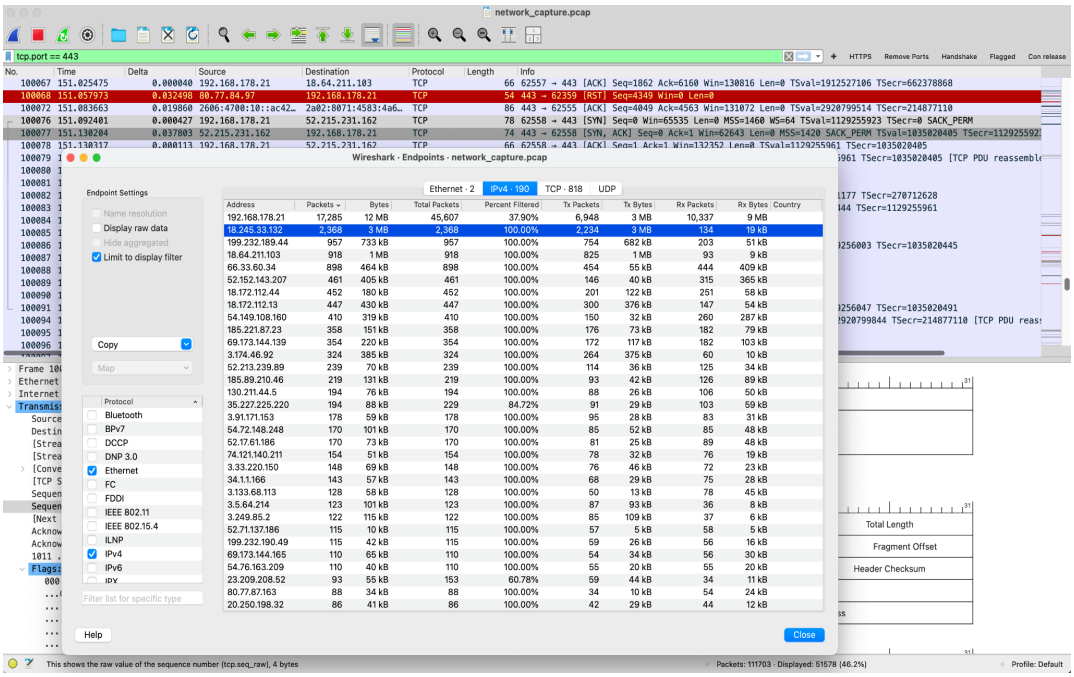
User Datagram Protocol

Source Address (ip.src), 4 bytes

Packets: 111703 · Displayed: 3381 (3.0%)

Profile: Default





Conclusion

The analysis demonstrates the ability to capture and interpret real network traffic using Wireshark. The findings show normal user activity with no strong indicators of malicious behavior, highlighting foundational SOC analyst skills.